Cereal 4 — Configs

Oct 26, 2026 · Anas Alsirafy

Both maize pipelines use a ConvNeXt-Base backbone trained in three phases. They differ in how the 8 labels are split across models. Values not listed here are in each notebook's `CONFIG` cell.

Monolithic model (`maize_multilabel.ipynb`)

Setting	Value
Backbone	ConvNeXt-Base, ImageNet weights
Head	Global average pooling → LayerNorm → Dropout 0.4 → Dense 256 ReLU → Dropout 0.3 → Dense 8 sigmoid
Pre-processing	CLAHE (L channel, clip 2.0, 8×8) + ImageNet normalisation
Augmentation	RandomResizedCrop, horizontal flip, vertical flip, transpose, colour jitter, Gaussian blur
Loss	Weighted BCE; pos_weight = (N − positives) / positives, capped at 50
Phase 1	Backbone frozen, lr 1e-4, 15 epochs
Phase 2A	Top 30 backbone layers unfrozen, lr 1e-5, clipnorm 1.0
Phase 2B	Whole network, lr 3e-6
Folds	5-fold MultilabelStratifiedKFold
Thresholds	Per label, grid 0.20–0.80 in steps of 0.05, chosen on OOF predictions

Specialist pipeline (`maize_specialist.ipynb`)

Five models run in parallel; their outputs are combined into the 8 labels. There is no gatekeeper model.

Model	Task	Output	Training images	Folds
S1	Healthy vs GLS	1 sigmoid	NoFoliarSymptoms or GLS	StratifiedKFold, 5
S2	Healthy vs NCLB	1 sigmoid	NoFoliarSymptoms or NCLB	StratifiedKFold, 5
S3	Healthy vs PLS	1 sigmoid	NoFoliarSymptoms or PLS	StratifiedKFold, 5
S4	Healthy vs CR	1 sigmoid	NoFoliarSymptoms or CR	StratifiedKFold, 5
S5	SR, NoFoliarSymptoms, Other, UnidentifiedDisease	4 sigmoids	All 2,355	MultilabelStratifiedKFold, 5

Shared training settings (`train_specialist()`):

Setting	Phases 1 and 2A	Phase 2B
Monitored metric	Validation AUC (max)	Validation AUC (max)

Setting	Phases 1 and 2A	Phase 2B
ReduceLROnPlateau	factor 0.3, patience 3, min lr 1e-7	factor 0.2, patience 2, min lr 1e-8
Early stopping	patience 5, restore best weights	patience 5, restore best weights
Optimiser	Adam (2A with clipnorm 1.0)	Adam, clipnorm 1.0
Learning rates	`lr_phase1`, `lr_phase2a` from CONFIG	`lr_phase2b` from CONFIG

Latest change: S5 no longer fine-tunes. Its frozen phase reached validation AUC 0.8026; unfreezing gave 0.7830 (top 30 layers) and 0.7919 (full network).